

Ethan Baek

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EDUCATION

University of California, Irvine

Irvine, CA

Bachelor of Science in Computer Science

Sep. 2023 – June 2027

Relevant Coursework: Data Structures & Algorithms, C++ Programming, Operating Systems, Programming in Assembly, Linear Algebra

University of Carlos 3 Madrid

Madrid, Spain

UCAEP Study Abroad

Aug. 2024 – Jan. 2025

EXPERIENCE

AI Fellowship

July 2024 – Sep. 2024

Headstarter AI

Remote

- Created 5 AI-driven projects (chatbots, vision, and recommendation tools) to solve real world tasks
- Built and iterated on LLM features using OpenAI APIs, improving response relevancy by 50%
- Implemented full-stack features in React and Node.js/Express with sub-200ms API latency in testing

Inventory Manager

June 2024 – Sept. 2024

Realself.

Seattle, WA

- Managed inventory of 200+ pieces of technical equipment, maintaining 100% tracking accuracy
- Reduced equipment retrieval time by 30–40% by reorganizing labeling and storage system

PROJECTS

Music Web | React.js, Node.js, OAuth 2.0, HTML/CSS

July 2025 – Present

- Built a full-stack Next.js app with OAuth 2.0 + Spotify APIs, enabling users to explore their top artists and 6+ related artists via an interactive node map
- Optimized Spotify and Last.fm integrations by removing deprecated endpoints and cutting redundant calls, keeping artist graph loads under 300ms

Video Hand Controller | Flask, React, OpenCV, NumPy, MediaPipe, TensorFlow

Apr. 2024 – Jan. 2025

- Developed a Chrome extension + Flask REST API to control video playback via hand gestures, reducing reliance on keyboard shortcuts
- Achieved real-time hand tracking at 30 FPS with 95% gesture accuracy in normal lighting
- Optimized MediaPipe + TensorFlow + NumPy pipeline to cut per-frame latency by 8ms

Personal Pokédex | Flask, React, SQLAlchemy, React Router

Jan. 2024 – June 2024

- Built a full-stack Pokédex using Flask, React, and a normalized SQL schema indexing 1,000+ Pokémon entries, enabling search and filtering with sub-100ms response times
- Added user profiles with saved teams and recent searches, reducing external API calls by 40% via caching and indexed lookups.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, C++, C, Java, Ruby

Frameworks: React, Next.js, Flask, Ruby on Rails, Node.js, Tailwind

Tools: Git, VS Code, RStudio, Microsoft Office

Libraries: NumPy, TensorFlow, Leaflet

Databases: SQL

ADDITIONAL EXPERIENCE

Philanthropy Chair

Sept 2024 – June 2025

Sigma Chi - Eta Sigma

Irvine, CA

- Led 4 major philanthropy events, coordinating 125+ members and alumni, raising \$90,000+ for the Huntsman Cancer Institute